Course Title: Software Engineering
Course: CSE412
Section:05

Title: Software Requirements Specification (SRS) for LearnSpace: An E-Learning Platform

Submitted to-

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1. Introduction

1.1 Project Overview

With the rapid growth of digital education and the increasing demand for flexible learning opportunities, online learning platforms have become an essential part of modern education systems.The system lets students look at and sign up for courses. They can do lessons that are laid out for them to take part in quizzes and see how they are doing. At the time instructors who have been verified can make courses add educational material see what students are doing and get paid when students sign up for their courses. The system also helps the people in charge make sure everything runs smoothly. It does this by checking that instructors are who they say are looking at the courses and keeping an eye on money transactions. The system is like a helper for students, instructors and the people in charge making sure the learning platform is working well for the students and the courses and the instructors. This comprehensive learning ecosystem is developed under the name "LearnSpace", aiming to bridge the gap between learning and earning through technology.

LearnSpace is an online learning system which was developed by instructors who know what other instructors need to build an open, scalable, and easy-to-use digital learning space.Our system lets any approved instructor build and teach courses, and students can sign up and pay right away to get to their lessons from anywhere.LearnSphere is different from your typical learning sites. It's built like a marketplace, which means instructors can make money sharing what they know and the skills they have.

From a learner's perspective:

> *"We want an effective and user-friendly platform to learn, not just academic things but an open platform to acquire knowledge"*

From an instructor's perspective:

> *"Many students want my course, but I cannot manage it as in Bangladesh, there is no open-source platform for us to help the students to learn also made an opportunity for us to get revenue by teaching online"*

These user stories define the functional backbone of LearnSphere and ensure that the system meets real-world requirements.

1.2 Purpose and Scope

The purpose of this project is to design and implement a learning platform by applying fundamental and advanced software engineering principles. The project focuses on demonstrating object-oriented programming, layered architecture, role-based access control, and secure online transaction handling within a real-world application context. The scope of the system includes user registration and authentication, instructor verification, course creation and management, online payment processing, learning progress tracking, and automated certificate generation. In addition, the system provides administrative functionalities such as user management, content approval, transaction oversight, and report generation. While the platform is developed as an academic project, its architecture and feature set are designed to be scalable and adaptable for practical deployment.

The scope of the system includes:

1. User registration and authentication with role-based access for example as Student, instructor or admin.
2. Instructor verification and course creation system.
3. Secure online payment for student and revenue-sharing mechanism for instructors
4. Course enrollment, learning progress tracking and certification facility.
5. Administrative control for moderation, reporting, and transaction monitoring of the system.

The system does not aim to replace large commercial platforms but rather to provide a scalable prototype suitable for local and global deployment.

1.3 Stakeholders

The LearnSpace platform has a number of key stakeholders whose interests and roles affect the functionality of the system. Each of the stakeholders has a different role to ensure the effective functioning of the system.

I. **Learners:** Students or learners are the end users who consume learning content, enroll in courses, and gain skills. They are the primary end users of the system and represent the largest stakeholder group.
II. **Instructors:** Instructors are the content creators here who design courses, teach learners, and earn revenue.
III. **Administrator:** Administrator panel is responsible for the governing and maintaining the whole platform.
IV. **Payment Gateway:** External payment gateways serve as supporting stakeholders by providing secure and reliable transaction services.

2. Engineering Process requirements

2.1 Stakeholder Needs & Analysis

The demands and expectations of the Learn-Space platform's core and secondary stakeholders were examined to determine the platform's requirements. Making sure the system is both technically possible and user-friendly while addressing real-world issues was the aim of this analysis. Several methods of requirement elicitation were taken into consideration to collect and evaluate requirements. To learn about prevalent issues with current learning platforms, informal interviews with instructors and students were carried out. Usability expectations, desired features, and payment-related issues were determined through online questionnaires and surveys. Additionally, feature gaps and areas for improvement were found through a comparative study of current platforms, such as local and worldwide learning systems. These techniques ensured that the criteria were realistic, user-focused, and consistent with actual use cases. These are the requirements that have been gathered.

Primary stakeholders: The primary stakeholders include students, instructors, and administrators.

1. Learner's Requirements

1) To provide easy navigation and usage, a user interface must be responsive and intuitive.

2) To make it easier for students to find pertinent courses, advanced course browsing and search features should be offered.

3) To safeguard user information and access, secure registration and authentication procedures must be put in place.

4) To enable secure and easy course enrollment, a dependable online payment method is necessary.

5) Following successful payment verification, registered students should have instant access to course materials.

6) Interactive tests, downloadable educational resources, and video lectures should all be supported by the platform.

7) To assist students in keeping track of their course completion, real-time progress monitoring must be made available.

8) After a course is successfully completed, certificates should be automatically created.

9) To improve accessibility, multilingual help should be offered, including Bangla and English.

10) The system needs to function consistently on a variety of gadgets.

2. Instructor (Teacher) Requirements

1) Before approving course publishing, a formal registration and verification procedure had to be implemented.
2) To enable autonomous course creation, updating, and maintenance, comprehensive course management tools should be made available.
3) It must be possible to submit papers, tests, and video content.
4) To track student involvement, open enrollment data should be accessible.
5) For optimal earning transparency, accurate and fast revenue computation is necessary.
6) To assess student involvement and course achievement, analytical dashboards must be made available.
7) Within the limitations set by the platform, flexible price choices must be offered.
8) To encourage ongoing development, access to student evaluations, ratings, and comments should be made available.

3. Administrator Requirements

1) Access permissions and user responsibilities must be controlled centrally.
2) It should be possible to verify the credentials of instructors and approve courses.
3) Financial transaction management and monitoring ought to be encouraged.
4) System use, performance, and revenue analysis reporting elements must be accessible.
5) There should be procedures for managing complaints, refunds, and instances of misuse.

Enforcing strict security measures is necessary to safeguard data privacy and system integrity.

Secondary stakeholders: Cloud service providers and outside payment gateway providers are examples of secondary stakeholders. While cloud services satisfy the platform's storage and scalability needs, payment gateways need secure integration points to complete transactions.

1. Payment Gateway Requirements

1) Transaction processing requires the usage of established and secure communication protocols.
2) Proper support should be provided for callback and transaction validation techniques.
3) It is necessary to guarantee accurate transmission of transaction data and payment status.
4) It is essential to adhere to industry-standard security and privacy regulations.

2.2 List of Requirements

2.2.1 Functional Requirements (FRs)

The functional requirements define what the system must do. The key functional requirements of LearnSpace are listed below:

Feature	Description
User Registration & Login	Allow students, instructors, and administrators to securely register and authenticate using role-based access control.
Role Management	Defines and manages user roles such as Student, Instructor, and Admin with appropriate permissions.
Instructor Verification	Enables administrators to verify instructor credentials before allowing course publication.
Course Creation	Allow verified instructors to create, edit, and publish courses with structured content.
Content Upload	Supports uploading of video lectures, documents, and learning resources by instructors.
Course Browsing & Search	Enables students to browse, search, and view detailed information about available courses.
Course Enrollment	Allow students to enroll in selected courses after successful payment verification.
Online Payment Processing	Integrates a secure payment gateway to handle course purchase transactions.
Transaction Verification	Verifies payment status before granting access to paid course content.

Learning Progress Tracking	Tracks student activity, completed lessons, and overall course progress.
Quiz & Assessment	Provides quizzes and assessments to evaluate student understanding.
Certificate Generation	Automatically generates certificates upon successful course completion.
Instructor Earnings	Calculates and displays instructor earnings based on course enrollments.
Reporting & Analytics	Generate reports related to users, courses, transactions, and platform performance.

2.2.2 Non-Functional Requirements (NFRs)

Non-functional requirements describe how the system performs rather than what it does.

Feature	**Description**
Security	Ensures data protection using encrypted passwords, secure authentication, and role-based access control.
Performance	Maintains responsive system behavior under concurrent user access.
Scalability	Supports future growth in users, courses, and transactions without major architectural changes.
Availability	Ensures the platform remains accessible with minimal downtime.
Usability	Provides a clean, intuitive, and user-friendly interface for all user roles.
Reliability	Guarantees accurate transaction handling and consistent data storage.

Maintainability	Follows modular and object-oriented design principles to simplify maintenance and upgrades.
Compatibility	Supports access across multiple devices and modern web browsers.
Localization	Supports multi-language content, including Bangla and English.
Integration	Ensures reliable communication with external services such as payment gateways and cloud storage.

2.2.3 Extra-Ordinary Requirements (wow factors)

The following features enhance the system beyond basic LMS functionality:

1.The system provides intelligent learning pathways that recommend subsequent courses based on a learner's progress and completed content, enabling structured and goal-oriented learning.

2.The platform supports an open marketplace model where verified instructors can independently publish courses and earn revenue through a transparent and automated revenue-sharing mechanism.

3.To improve accessibility and inclusiveness, the system supports multilingual content delivery, including Bangla and English, allowing users to learn in their preferred language.

4. The system offers a comprehensive skill dashboard that visually displays completed courses, earned certificates, and overall learning progress, helping learners track their development effectively.

5.The platform enables the creation of short, focused courses designed for quick skill acquisition, catering to learners with limited time or specific learning goals.

6.Integrated discussion forums like live chatbot and course-level interaction spaces encourage communication between learners and instructors, fostering collaborative learning and knowledge sharing.

2.3 House of Quality (QFD Integration)

Quality Function Deployment (QFD) is a structured approach used to translate customer needs into technical and engineering requirements. In the LearnSpace platform, QFD is applied to ensure that stakeholder expectations are systematically mapped to appropriate technical solutions. The House of Quality matrix helps identify the relationships between customer requirements and engineering characteristics, ensuring that system design decisions directly support user needs and quality objectives.

2.3.1 Customer Requirements (CRs) List

The following customer requirements were identified through stakeholder analysis and requirement elicitation techniques such as questionnaires and interviews:

I. CR1: Easy and intuitive platform navigation

II. CR2: Secure and trustworthy online payment system

III. CR3: High-quality and structured learning content

IV. CR4: Fair and transparent earning system for instructors

V. CR5: Fast and reliable system performance

VI. CR6: Clear progress tracking and certification

VII. CR7: Strong security and data protection

VIII. CR8: Multi-language support and accessibility

2.3.2 Engineering Requirements (TRs) List

To address the identified customer requirements, the following technical and engineering requirements were defined:

I. **TR1:** RESTful API-based system architecture

II. **TR2:** Role-based access control and JWT authentication

III. **TR3:** Secure payment gateway integration with transaction verification

IV. **TR4:** Scalable and well-structured database design

V. **TR5:** Modular backend architecture following MVC principles

VI. **TR6:** Responsive frontend design using modern frameworks

VII. **TR7:** Automated learning progress tracking and certificate generation

2.3.3 QFD Matrix (House of Quality)

The QFD matrix illustrates how customer requirements are translated into corresponding technical solutions. Strong relationships indicate direct and critical technical responses to user needs, while medium relationships represent supportive or indirect contributions. This structured mapping ensures that LearnSpace is designed to align closely with stakeholder expectations and quality objectives.

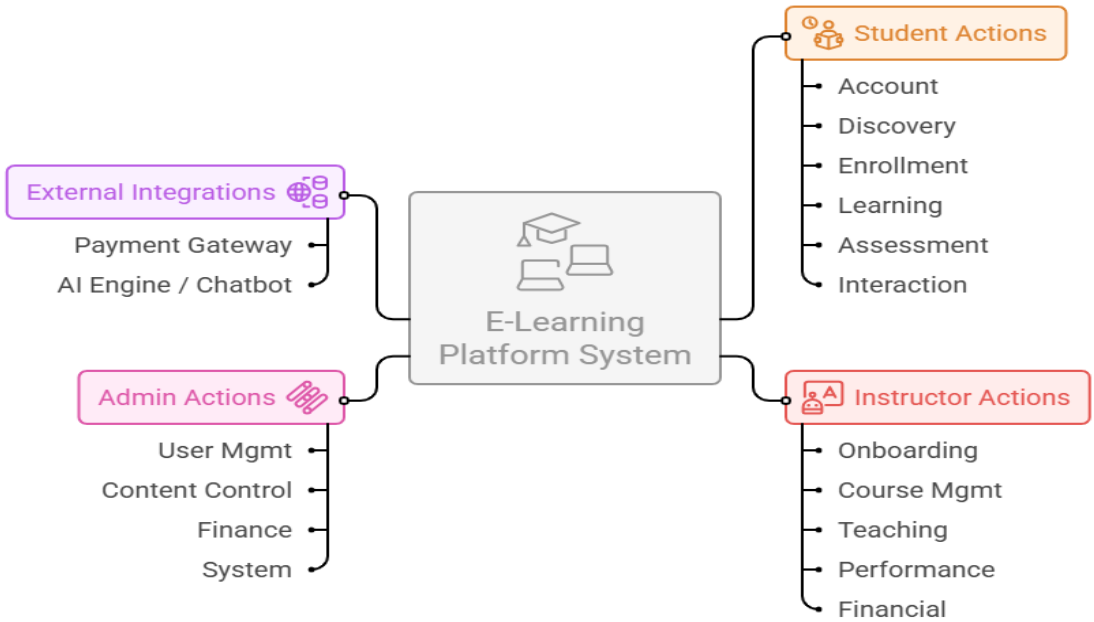
QFD Matrix (House of Quality) – LearnSpace

Customer Requirements (CRs)	**Technical Requirements (TRs)**	**Relationship**
CR1: Users need a simple and intuitive learning platform	TR1: Implement responsive UI using modern frontend framework	Strong
CR2: Users require secure login and access control	TR2: Implement JWT-based authentication and role-based access control	Strong
CR3: Students want safe and reliable online payments	TR3: Integrate secure payment gateway with transaction verification	Strong
CR4: Instructors want transparent and accurate earnings tracking	TR4: Implement automated revenue calculation and reporting module	Strong
CR5: Learners want to track progress and receive certificates	TR5: Implement automated progress tracking and certificate generation	Strong
CR6: Platform should support large number of users	TR6: Design scalable backend architecture and optimized database	Medium
CR7: Users expect high-quality learning content delivery	TR7: Implement structured course management and content upload system	Medium
CR8: Users want access in Bangla and English	TR8: Implement multi-language content and localization support	Medium
CR9: Administrators need control over courses and users	TR9: Implement admin dashboard for verification, moderation, and monitoring	Strong

CR10: Users expect the platform to be reliable and always available	TR10: Implement error handling, logging, and reliable server configuration	Medium

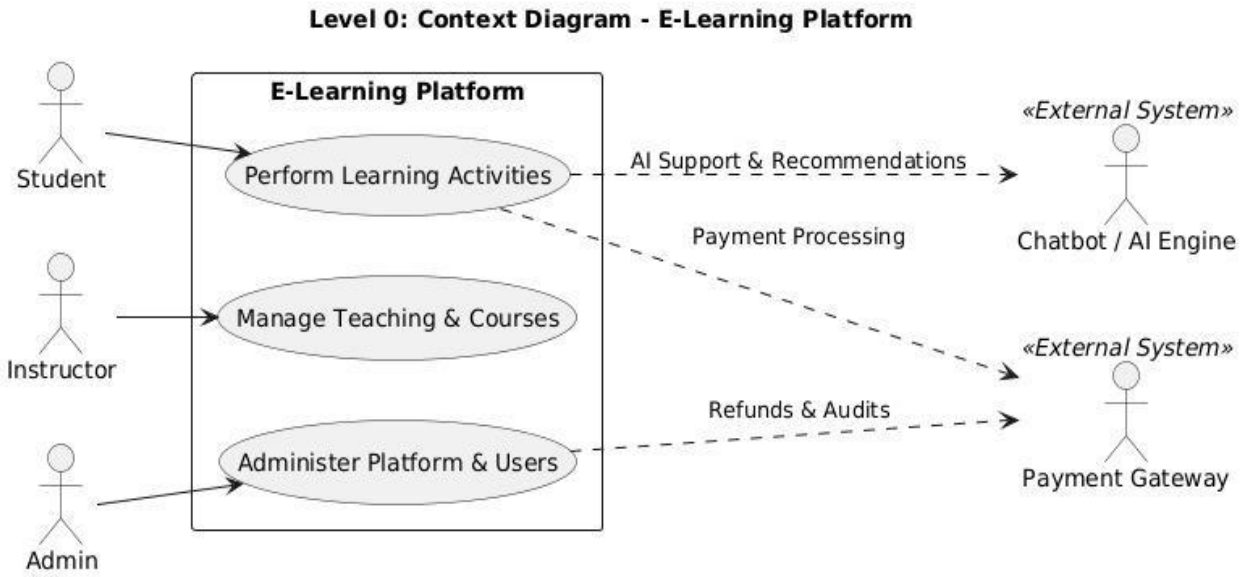
3. Diagrams

System overview of the system:

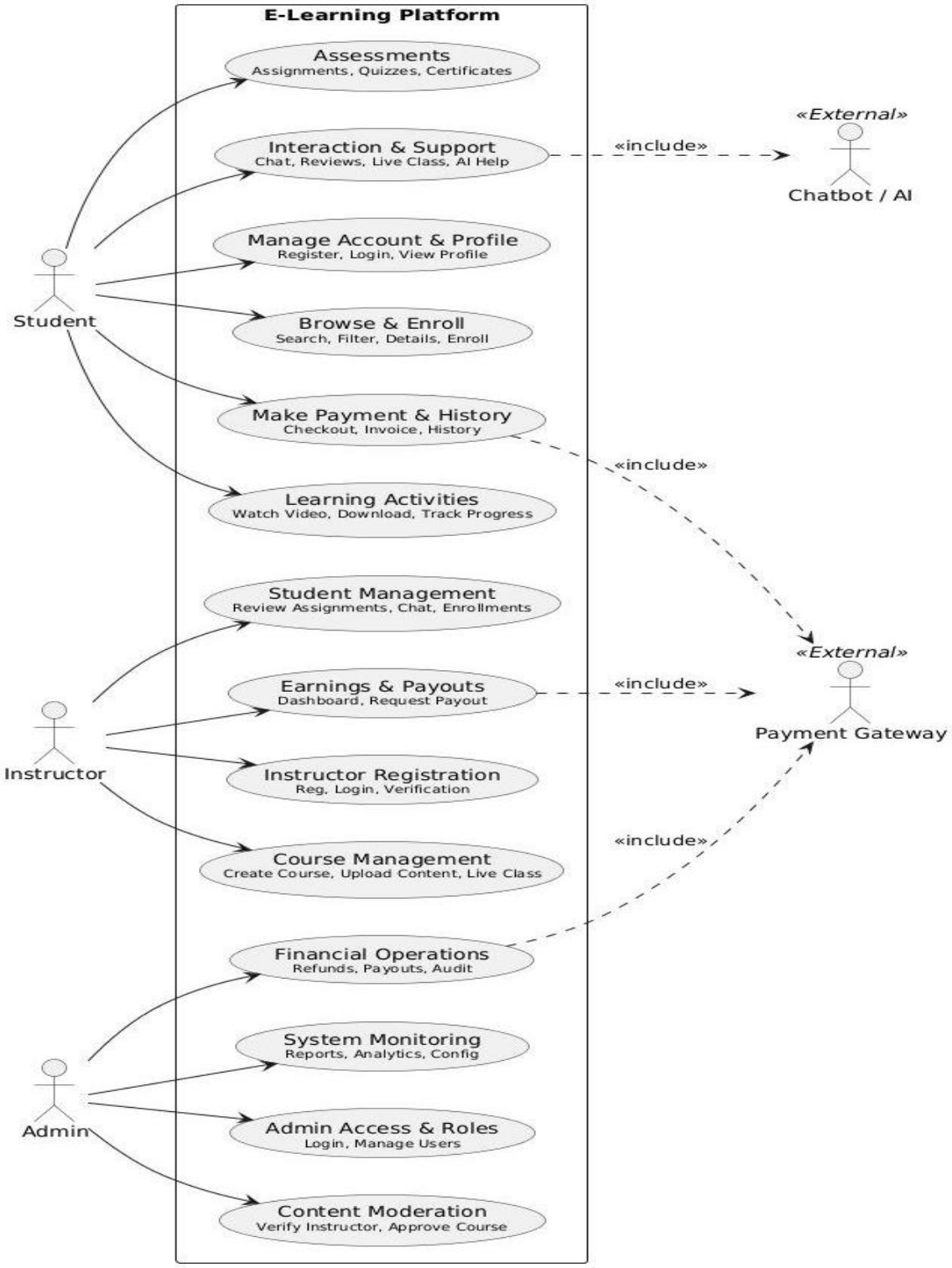


3.1. Use case diagram

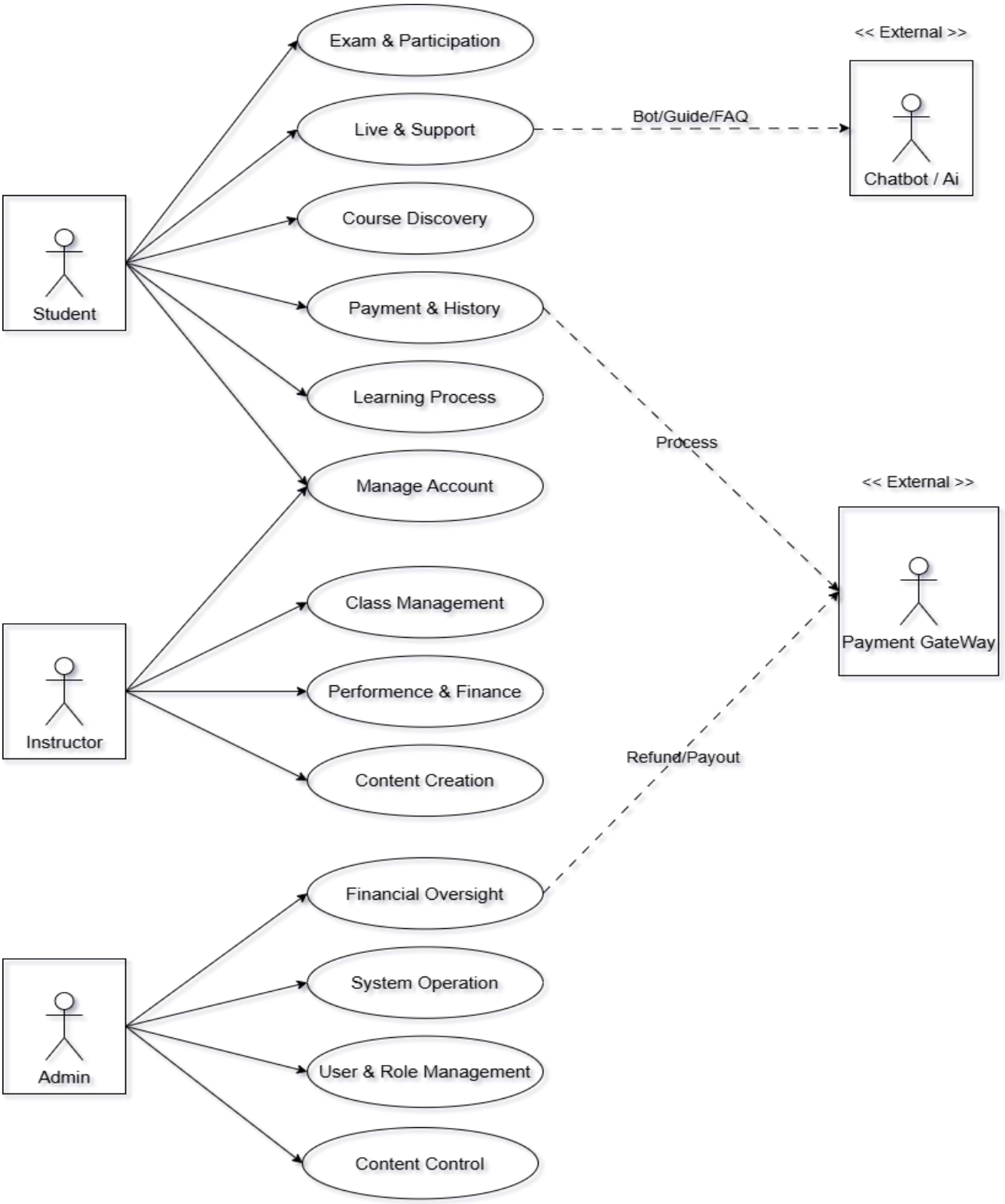
3.1 .1 Use case diagram level 0



3.1.2 Use case diagram level 1

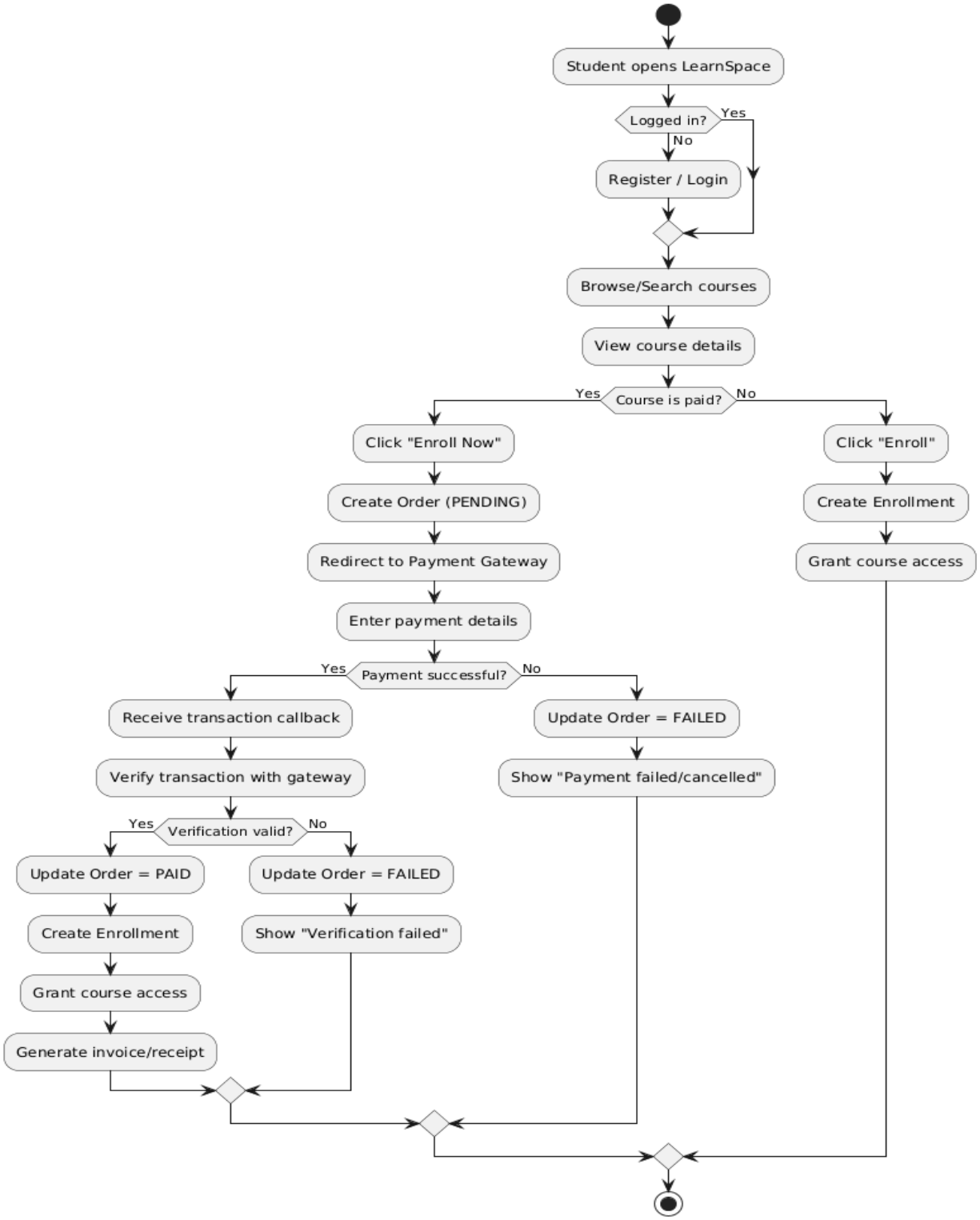


3.1.3 Use case diagram level 2

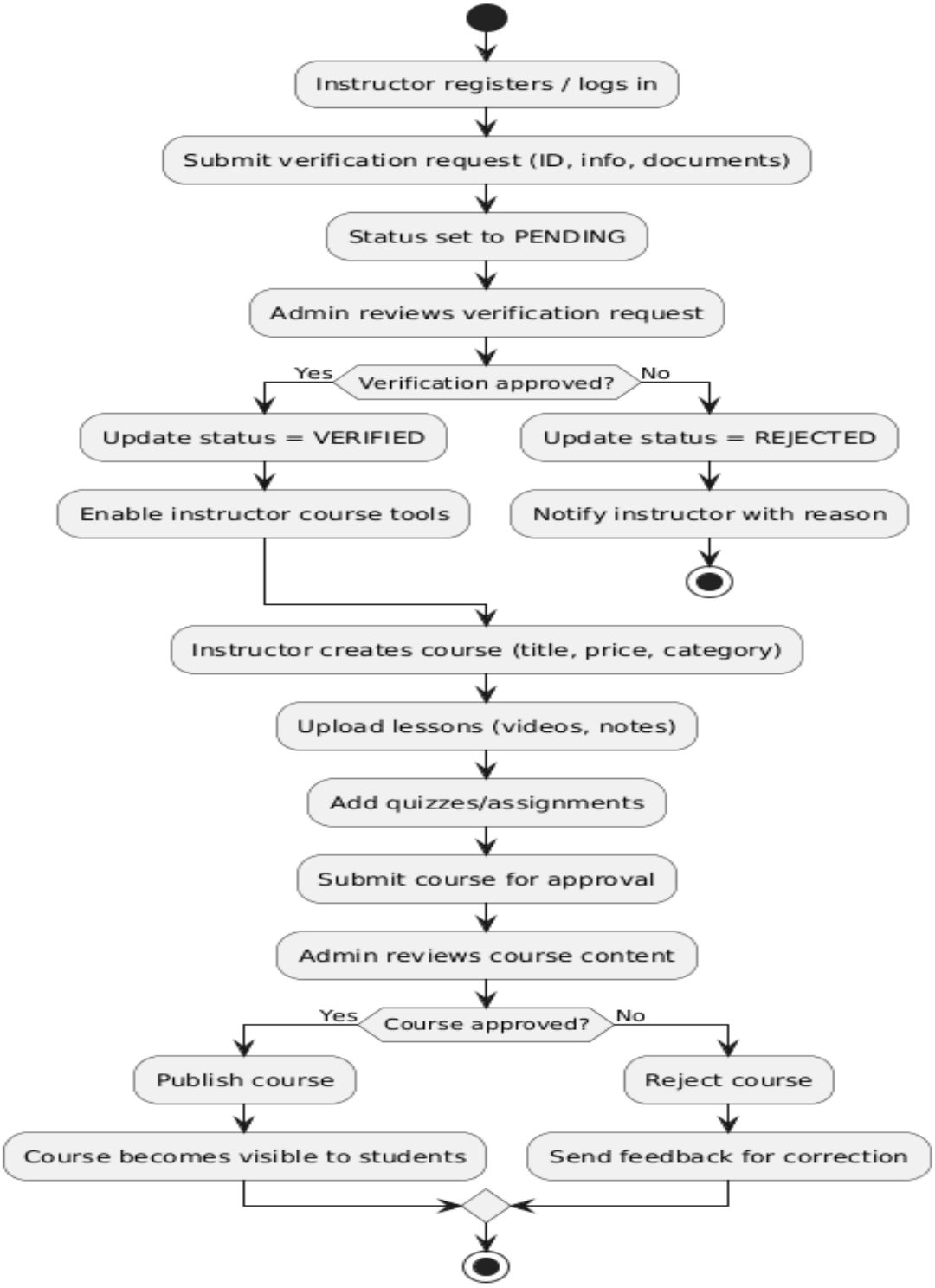


3.2 Activity Diagram

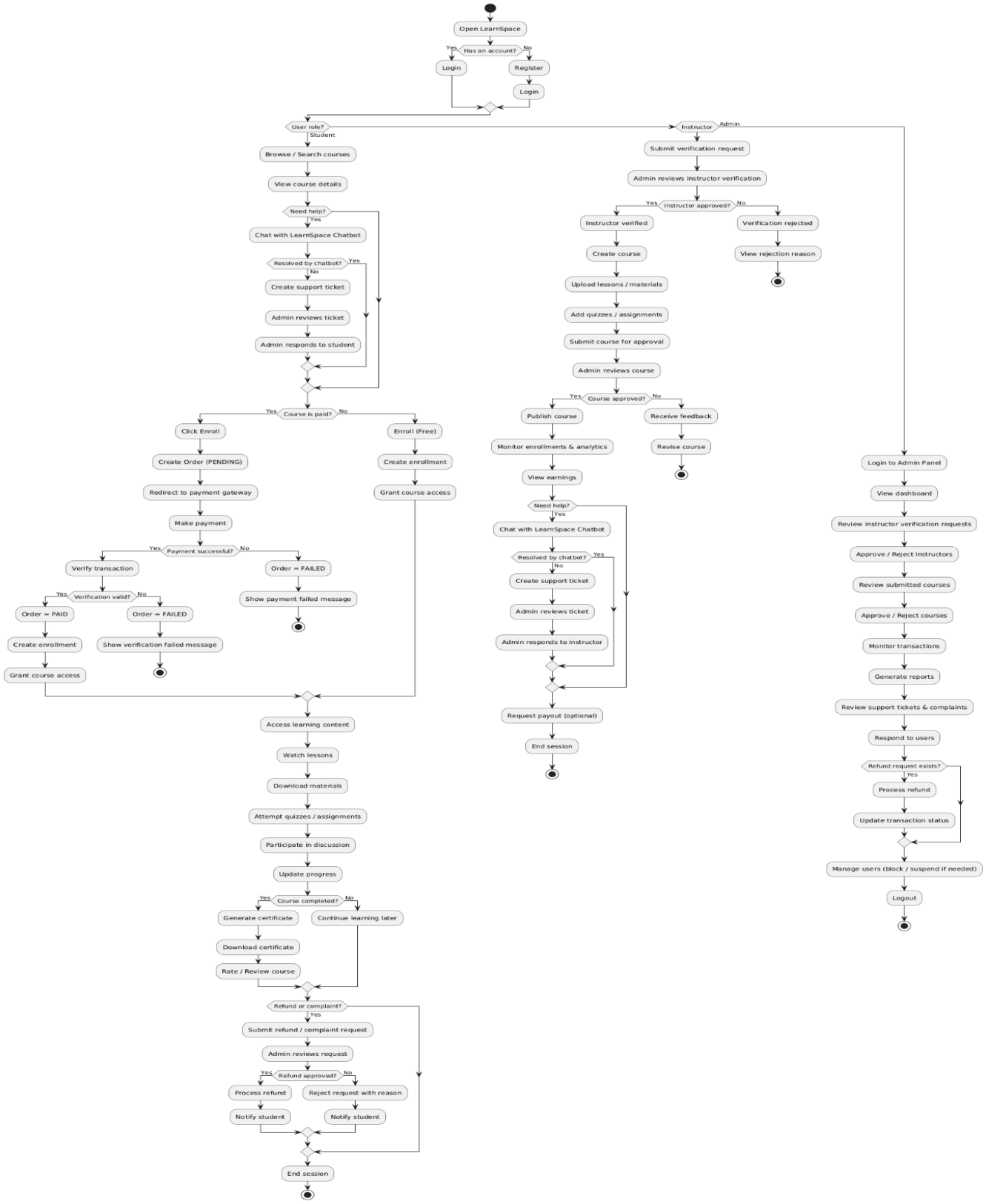
3.2.1 Activity Diagram: Course Enrollment and Payment Flow



3.2.2 Activity Diagram: Instructor Verification and Course Publishing Flow

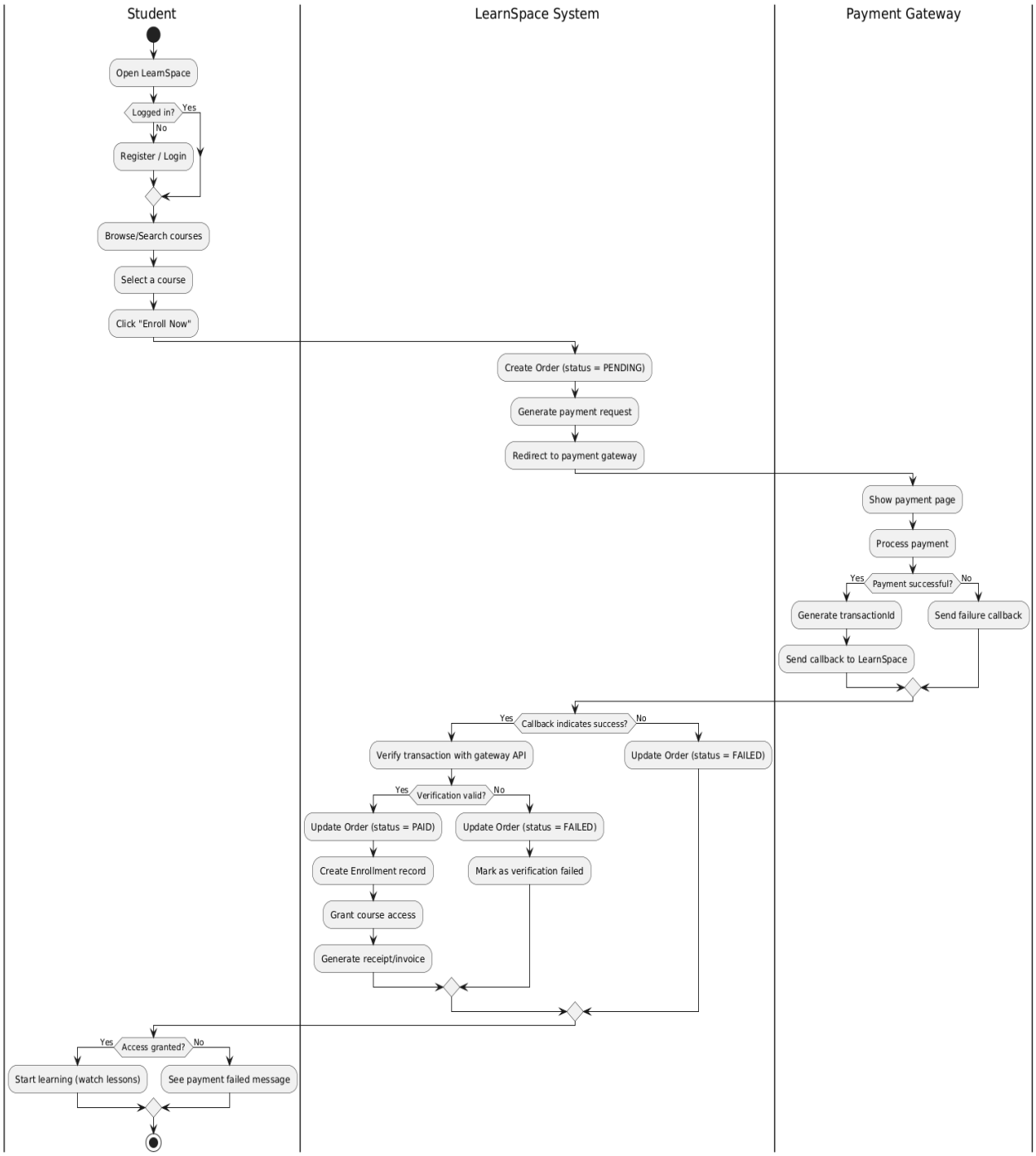


3.2.3 Activity Diagram of the system:

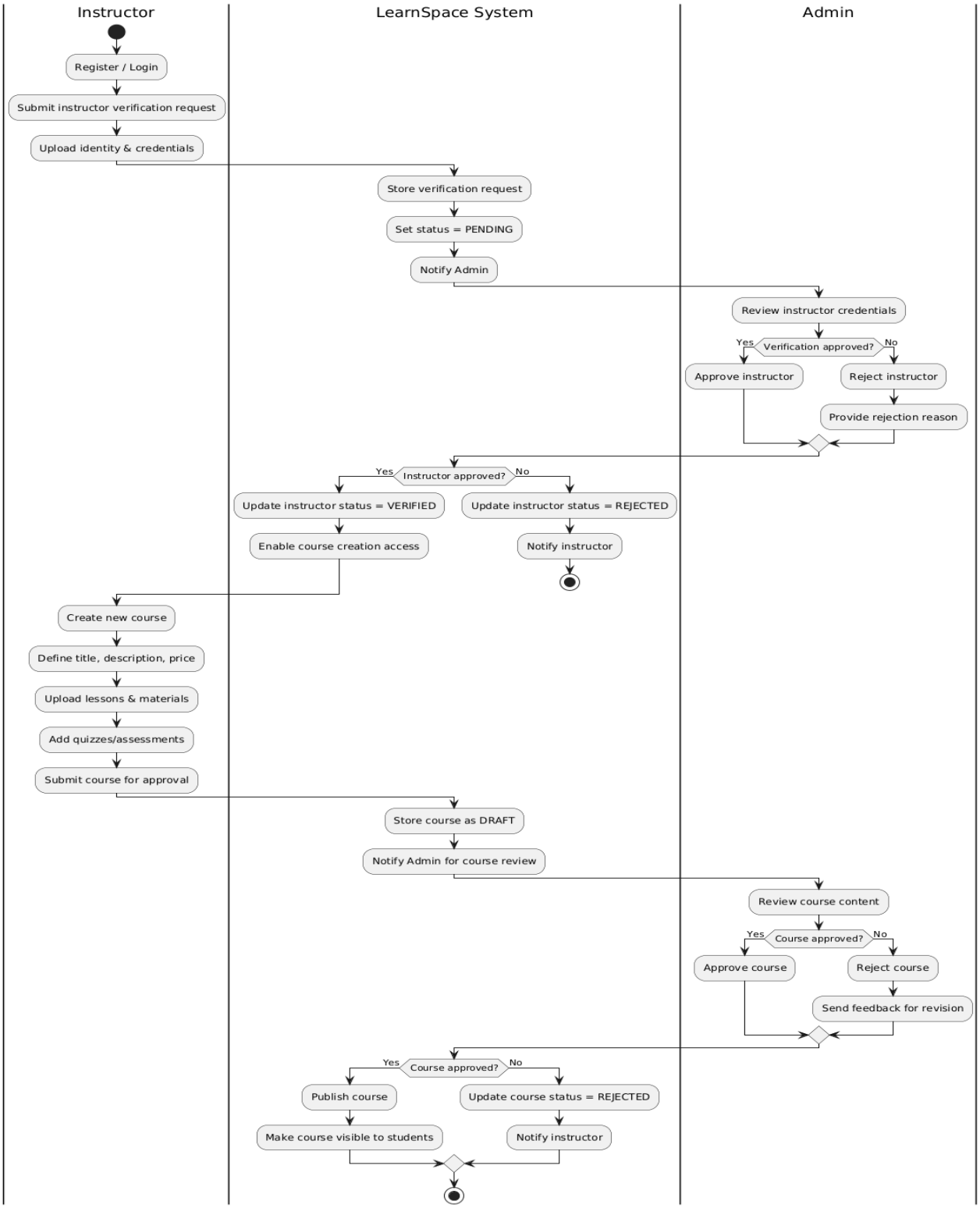


3.3 Swimlane diagram

3.3.1 Swimlane diagram for course enrollment and payment system

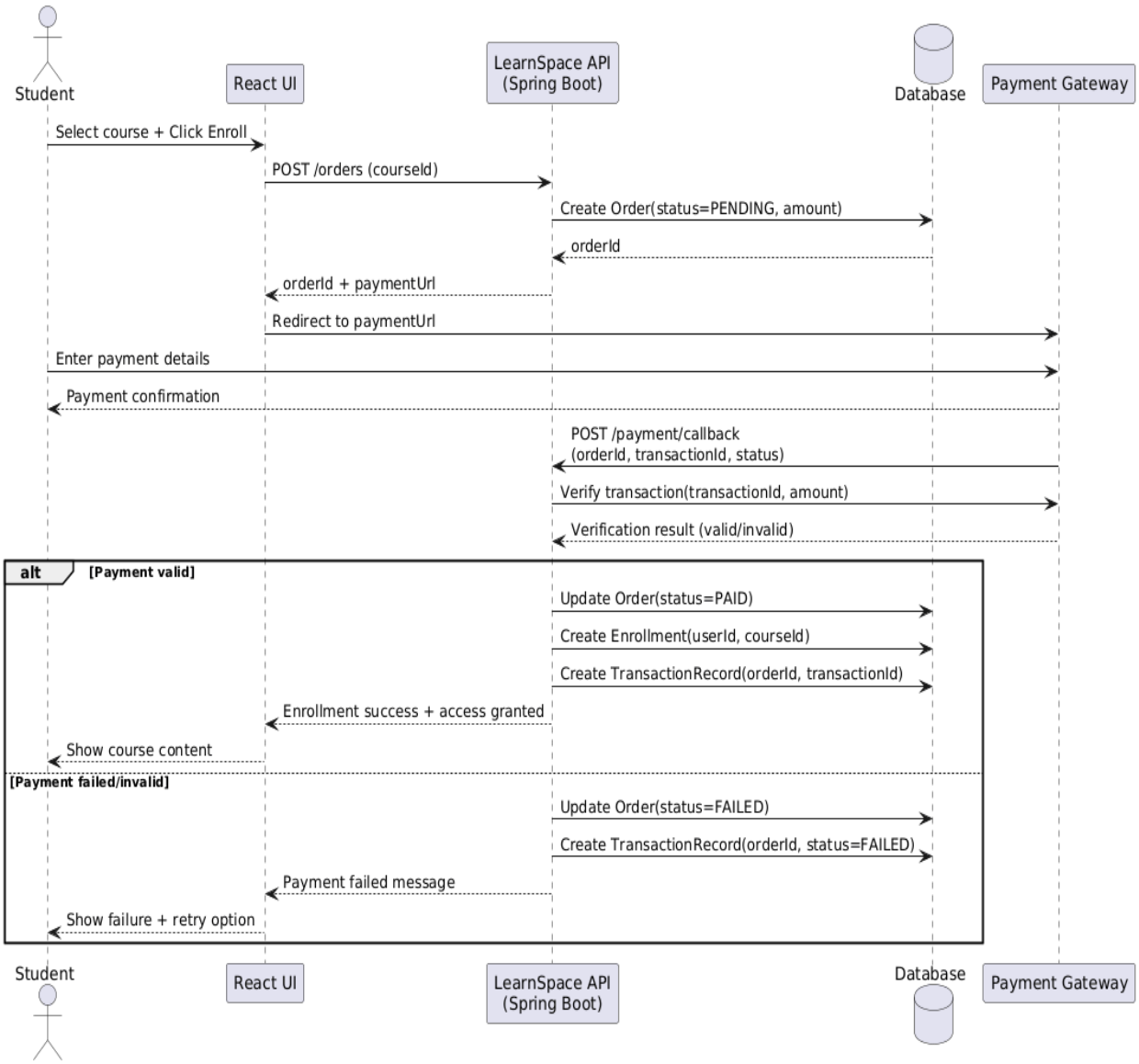


3.3.2 Swimlane Activity Diagram: Instructor Verification & Course Publishing

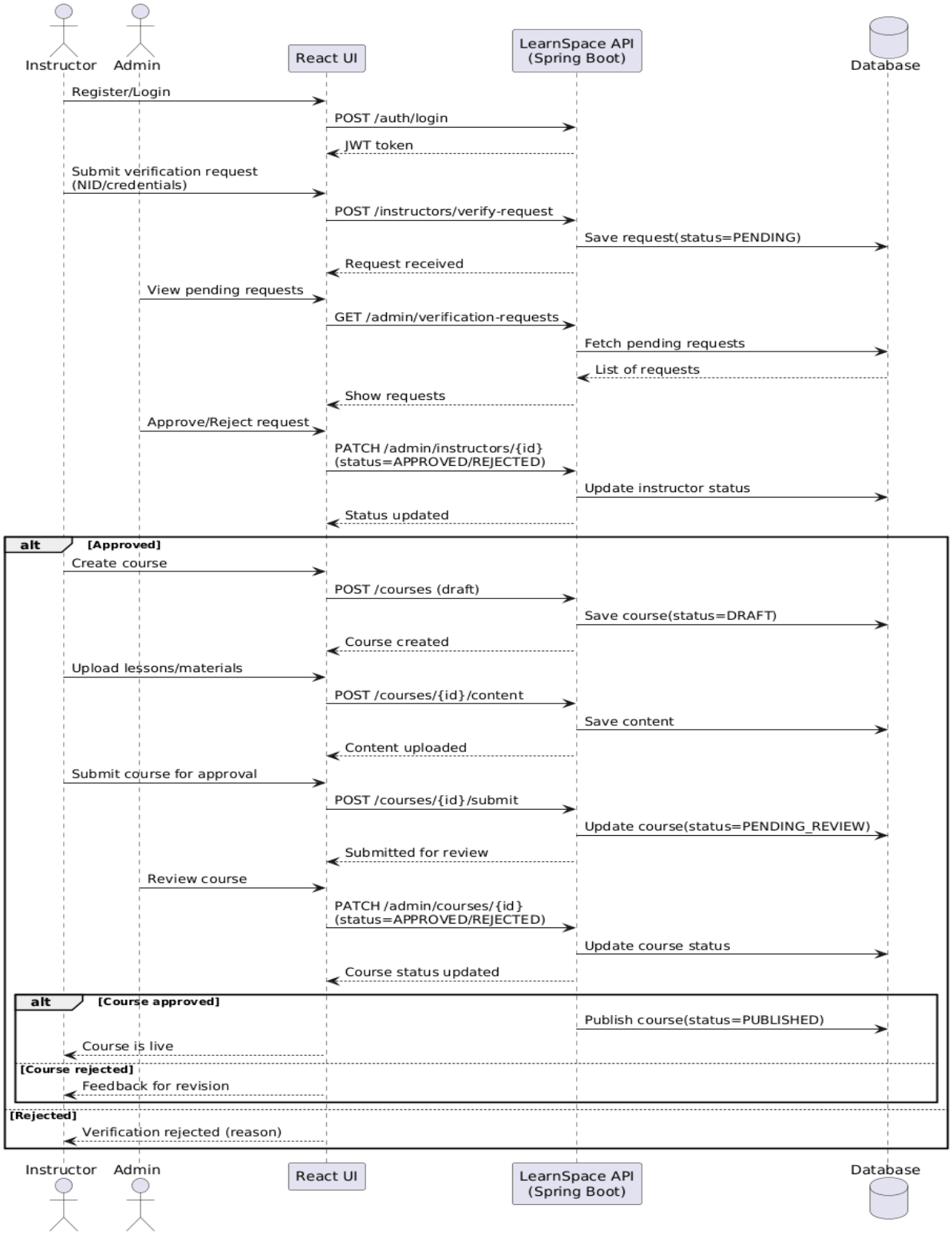


3.4 Sequence Diagram

3.4.1 Sequence Diagram: Course Enrollment and Payment Verification

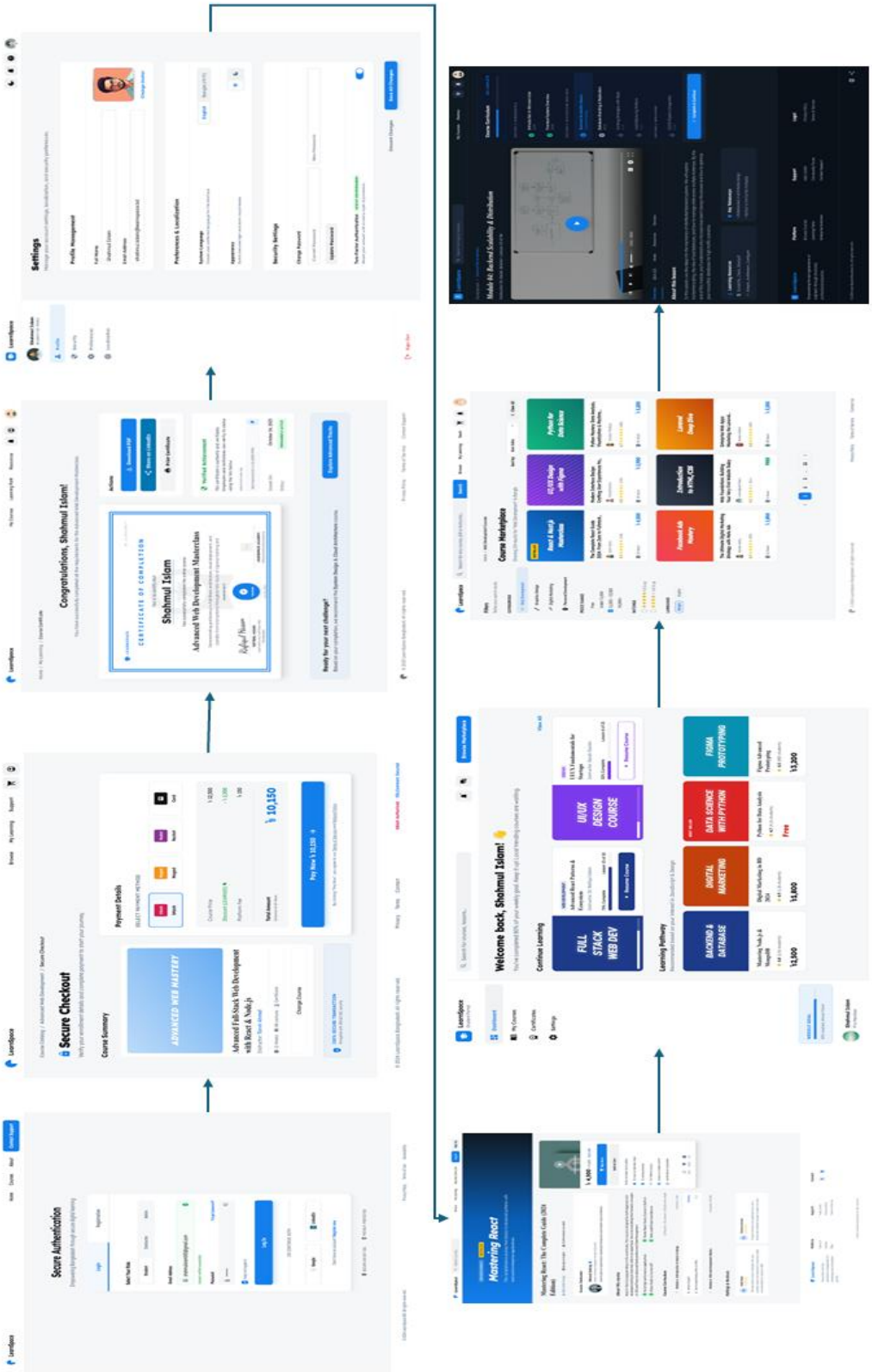


3.4.2 Sequence Diagram: Instructor verification and course publishing

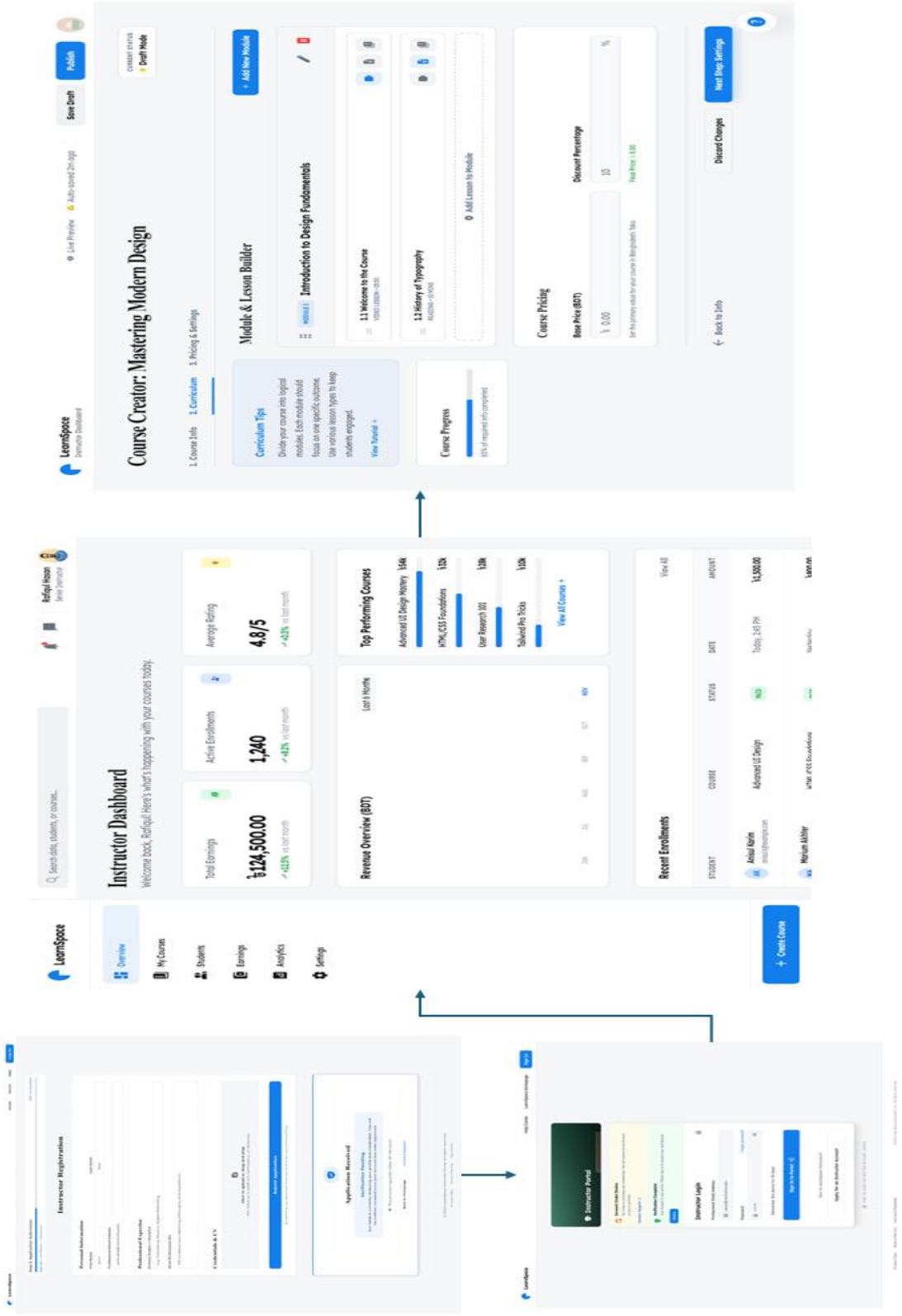


3.4 Prototyping using UI sketches

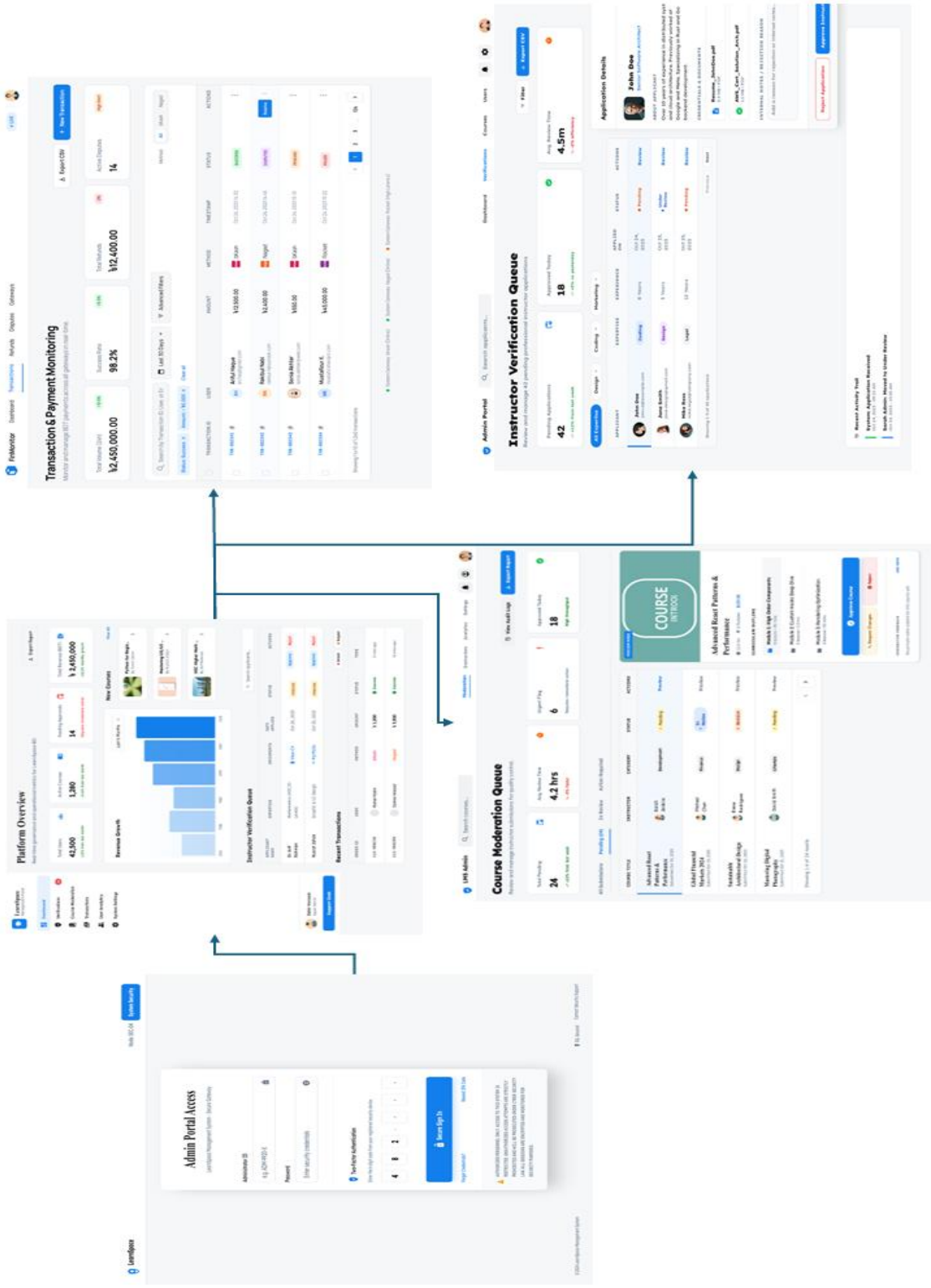
Student module:



Instructor module:



Admin module:



Conclusion

LearnSpace is designed as a complete and practical online learning platform that connects students, instructors, and administrators in a single system. The platform allows students to easily find courses, enroll, make payments, learn through structured content, and earn certificates. At the same time, verified instructors can create courses, share knowledge, and earn income, while administrators ensure that the system runs smoothly, securely, and fairly.This project follows proper software engineering principles by clearly identifying stakeholder needs, defining functional and non-functional requirements, and modeling the system using UML diagrams. Features such as secure payments, progress tracking, chatbot support, and instructor verification make the system more usable and realistic.

Overall, LearnSpace demonstrates how a modern learning platform can be built in a structured way, making it suitable both as an academic project and as a foundation for a real-world application.

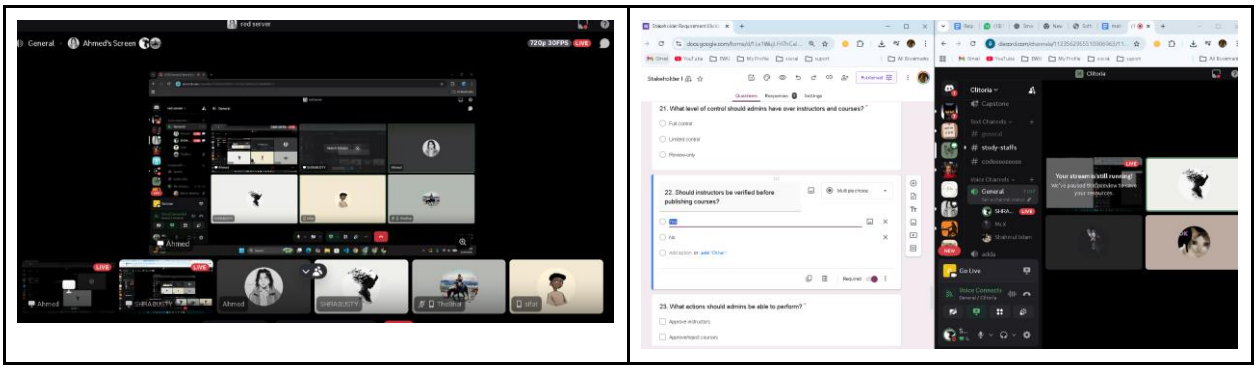
Requirement Elicitation Method

The requirements for the LearnSpace platform were collected using a combination of informal interviews and online questionnaires created through Google Forms. These methods were chosen to gather practical and user-focused requirements from different stakeholders in an efficient and flexible manner.

Informal interviews were conducted with students and instructors to understand their experiences with existing online learning platforms and the difficulties they commonly face. During these discussions, participants shared their expectations regarding course accessibility, ease of use, payment safety, learning support, and content management. The informal nature of the interviews helped participants speak freely and provide honest feedback, which was useful in identifying real problems and improvement opportunities.

In addition to interviews, Google Forms were used to collect structured responses from stakeholders. Separate questionnaires were designed for students, instructors, and administrators. The forms included both multiple-choice and open-ended questions to capture quantitative data as well as individual opinions. This approach made it easier to identify common requirements, prioritize features, and understand user preferences such as language support, payment methods, and system usability.

Link of google form-

https://docs.google.com/forms/d/e/1FAIpQLSeiQgWkUBQ3dNrMiBRM7SjeyuyLSdBfIGlI-XbsEkaGblTGbg/viewform?usp=header

Screenshots: